AP PRECALCULUS

Unit 1 • Lesson 1.7 practice • Answer key

Lesson 1.7 • Rational Functions and End Behavior

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

What is the horizontal asymptote of $r(x) = \dfrac{5x-1}{x^2+4}$?

(A) $y = 5$

(B) $y = 0$

(C) $y = -\dfrac{1}{4}$

(D) There is none.

Solution

Answer: (B). The degree of the numerator is less than the degree of the denominator.

Question 2

NO CALCULATOR

What is the horizontal asymptote of $r(x) = \dfrac{4x^3-x}{1-2x^3}$?

(A) $y = 2$

(B) $y = 4$

(C) $y = -2$

(D) $y = 0$

Solution

Answer: (C). Equal degrees: $\frac{4}{-2} = -2$.

Question 3

NO CALCULATOR

What is the slant asymptote of $r(x) = \dfrac{x^2+3x}{x-1}$?

(A) $y = x+3$

(B) $y = x+4$

(C) $y = x$

(D) $y = x-4$

Solution

Answer: (B). $x^2+3x = (x-1)(x+4)+4$, so $r(x) = x+4+\dfrac{4}{x-1}$.

Question 4

NO CALCULATOR

$\lim_{x\to-\infty} \dfrac{x^3}{x^2+1}$ is

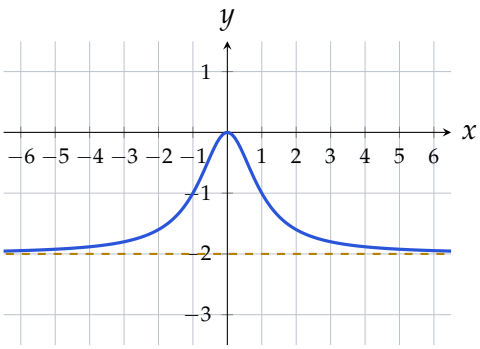
(A) 1
(B) 0
(C) $+\infty$
(D) $-\infty$

Solution
Answer: (D). The quotient behaves like $\frac{x^3}{x^2} = x$, which goes to $-\infty$.

Question 5 (graph)

NO CALCULATOR

The graph of $r(x) = \dfrac{-2x^2}{x^2+1}$ is shown. What is $\lim_{x\to\infty} r(x)$?



(A) 0
(B) -2
(C) $-\infty$
(D) 2

Solution
Answer: (B). The graph levels off toward the dashed line $y = -2$. Algebraically, equal degrees give $\frac{-2}{1} = -2$.

Question 6 (Free response)

NO CALCULATOR

Let $r(x) = \dfrac{6x^2+x}{3x^2-12}$ and $s(x) = \dfrac{2x^2-x+3}{x+1}$.

(a) Write limit statements for the end behavior of r.
(b) Find the equation of the slant asymptote of s.
(c) For large positive values of x, is the graph of s above or below its slant asymptote? Explain.

Solution
(a) Equal degrees: $\lim_{x\to\pm\infty} r(x) = \frac{6}{3} = 2$.
(b) Synthetic division by -1 on $2, -1, 3$ gives $2, -3$ and remainder 6: $s(x) = 2x - 3 + \dfrac{6}{x+1}$.
Slant asymptote: $y = 2x - 3$.
(c) For $x > -1$, $\dfrac{6}{x+1} > 0$, so $s(x) > 2x - 3$: the graph is above the asymptote.